

Energy and reactions

A LOOK AT THE WAY ENERGY IS INVOLVED IN CHEMICAL REACTIONS.

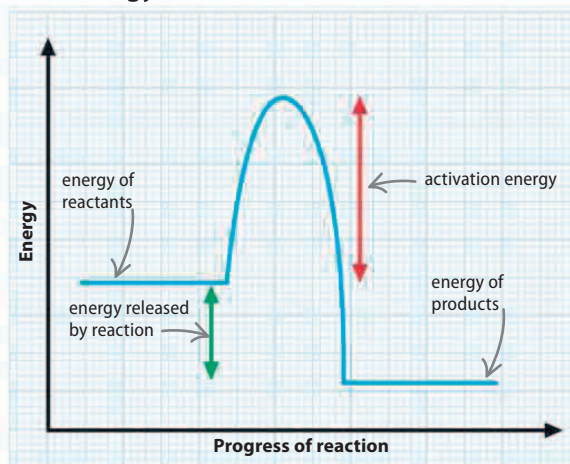
All chemical reactions require energy. Energy is needed to begin breaking and reforming atomic bonds. Most reactants need energy added to them before they will react.

Activation energy

The activation energy is the amount of energy that a reaction needs to begin. It is like a hill that the reactants have to get over. A reaction between a strong acid and alkali has low activation energy. It occurs as soon as the reactants are mixed because the molecules have enough energy already. The combustion of coal has a higher activation energy, so coal must be heated (adding energy) before it will burst into flames.

▷ Energy graph

When the energy involved in a reaction is shown as a graph, the activation energy forms a hump, over which the reactants must pass to form products.



Exothermic reaction

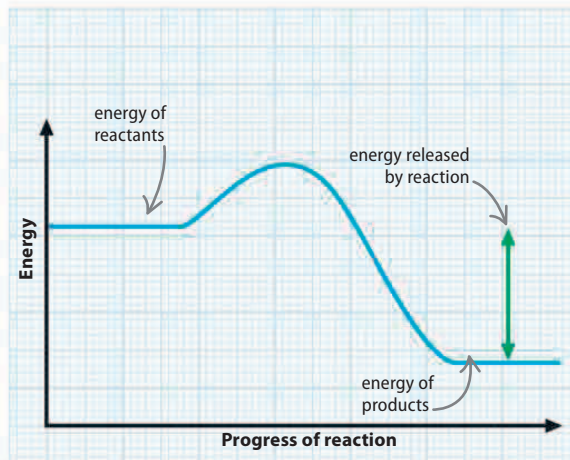
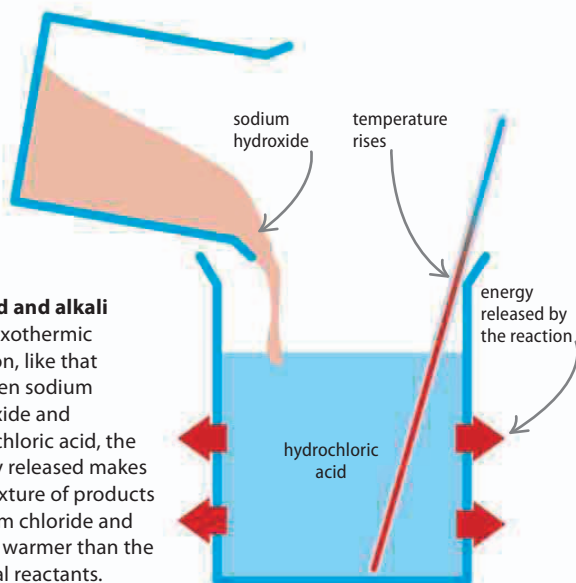
Chemical reactions need energy to begin, but they also release energy as the reactants reorganize into products. When the amount of energy released is greater than the activation energy, the reaction is exothermic. Exothermic reactions, such as the combustion of fuels, heat the surroundings with this release of energy.

▽ Energy released

In an exothermic reaction the energy in the products is lower than that in the reactants. This is because energy is lost as heat during the reaction.

▷ Acid and alkali

In an exothermic reaction, like that between sodium hydroxide and hydrochloric acid, the energy released makes the mixture of products (sodium chloride and water) warmer than the original reactants.



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◀ 130–131 Combustion

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Catalysts 138–139 ▶

Energy 170–171 ▶